# ZiraHive — Business Requirements & Software Requirements Specification

Oct 18, 2026 · @Mahbub

## 1. Document control

This document states what ZiraHive must do for its users (BRD, sections 2–6) and how the software meets it (SRS, sections 7–13), describing release 1.0 plus the work on the `develop` branch toward 1.1.

| Item | Value |
|---|---|
| Product | ZiraHive (formerly GharBox), self-hosted home cloud and media server |
| Owner | Mahbub (product owner, maintainer, first operator) |
| Version described | 1.0.0 (released, `main`) + 1.1 in progress (`develop`) |
| Audience | The owner, contributors, testers, and family members who run or review it |
| Licence | AGPL-3.0, free and open source |
| Related documents | `README.md`, `docs/FIRST_RUN.md`, `docs/REMOTE_ACCESS.md`, `docs/BACKUP.md`, `docs/IPTV.md`, `docs/AI.md`, `docs/PLUGINS.md`, `docs/ROADMAP.md` |

Requirement IDs are stable: **BR-nn** for business requirements, **FR-AREA-nn** for functional and **NFR-nn** for non-functional requirements. Priority uses MoSCoW (Must, Should, Could, Won't for now); status is Done (1.0), Done (develop) or Planned.

## 2. Executive summary and business context

ZiraHive turns a family's existing Windows PC into one private service for phone backup, files, photos and media streaming, reached from any browser at home or away, at no ongoing cost.

**The problem.** A typical household in Dhaka already owns a PC with several terabytes of disk, yet:

- phone storage fills up, and backing up means a monthly cloud plan that grows every year;
- films, series, music and books sit on the PC's drive, watchable only at that PC;
- photos are scattered across several phones with no shared family timeline;
- the self-hosted alternatives are separate tools (photo backup, file sync, media server), each with its own install, accounts and updates;
- remote access is hard because most Bangladeshi ISPs put homes behind carrier-grade NAT, so port forwarding is not an option.

**The answer.** One Docker stack on the home PC that combines a Google Drive-style backup and file store with a Jellyfin-style media server, plus watch parties, Live TV, an end-to-end encrypted vault and optional local AI. Clients are browsers only: no store apps. Remote access runs over Tailscale, which works behind CGNAT without opening a port.

**Why now.** Release 1.0 shipped all nine planned batches and is running on the owner's PC with real media (`G:\Media`). The `develop` branch adds Live TV, a photo vault with folders and an adjustable auto-lock, and moves toward 1.1. The next business question is whether ZiraHive can be installed by other families with only the setup script and the first-run guide.

## 3. Business objectives and success measures

Success means the family stops paying for cloud storage and uses ZiraHive daily, with no data loss and no unencrypted private data leaving the home.

| # | Objective | Success measure | Target |
|---|---|---|---|
| O1 | Replace paid cloud backup | Monthly spend on cloud storage | 0 BDT within 3 months of daily use |
| O2 | Back up every family phone | Share of family phones backing up to ZiraHive | All phones; no backup older than 7 days |
| O3 | Make the media library usable on any screen | Titles playable in a browser, directly or converted | 100% of scanned titles play on phone, PC and TV |

| # | Objective | Success measure | Target |
|---|---|---|---|
| O4 | Never lose data | Offsite backup success and read-back checks | Daily snapshot; weekly check passes; restore rehearsed once per release |
| O5 | Keep private data private | Private data reachable from the internet or stored unencrypted on the server | None: app only over Tailscale; vault content only as ciphertext |
| O6 | Easy to install | Time from download to first sign-in on a fresh Windows PC with Docker Desktop | Under 30 minutes, using only `setup.ps1` and `docs/FIRST_RUN.md` |
| O7 | Stay free and open | Features behind a paywall or needing a paid service | None (TMDB key and IPTV subscription stay optional, user-supplied) |
| O8 | Bangla-first | Interface texts with a Bangla translation | 100%, enforced by a CI check |

O6 has not yet been measured on a PC other than the owner's; it is the main open question before inviting other families.

## 4. Stakeholders and user personas

Five kinds of people use ZiraHive; the admin carries all the setup and safety work so that everyone else only needs a browser.

| Persona | Who | Needs | Access |
|---|---|---|---|
| Admin (owner) | The person who runs the PC, e.g. Mahbub | Install and update; add people, quotas and age limits; categories; remote access; backups; health and security checks | Full; still cannot open anyone's vault |

| Persona | Who | Needs | Access |
|---|---|---|---|
| Family member | Spouse, parents, siblings at home | Back up phone photos; find files; watch, listen, read; shared albums; watch parties | Own data, shared albums, categories allowed to them |
| Child profile | Children with an age limit (7+ or 13+) | Watch and read age-appropriate titles only | Titles rated within the limit; no Live TV |
| Remote family | Relatives in another city or abroad | Watch together, see shared albums | Over Tailscale (invited to the tailnet), or a linked ZiraHive server |
| Guest | Anyone given a share link | Open one file, folder or album | Share link only (optional password, expiry, view-only); no account |

Other stakeholders: **contributors** (code quality, tests, docs), **the IPTV provider** (connection limits, terms of service), **Tailscale** (free plan limits) and **TMDB** (optional poster and rating data).

## 5. Scope

ZiraHive 1.x is a single-household server on one Windows PC, used through browsers; anything that needs a store app, a hosted service or a second server is out of scope for now.

**In scope (1.x)**

- Accounts, roles, quotas, age limits, passkeys and two-factor sign-in
- Phone and PC backup (browser uploads, ZiraHive Sync for Windows), files, versions, 30-day trash, full-text search and OCR, document scanner
- Photos: timeline, albums, memories, sharing, clean-up, trash, photo vault
- End-to-end encrypted vault with folders and auto-lock
- Media from folders in place: movies, series, mixed, music, audiobooks, books, comics; conversion while playing; offline copies
- Live TV from a user-supplied Xtream Codes IPTV subscription
- Watch parties with chat, reactions and voice; linked ZiraHive servers
- TVs: Chromecast, DLNA (via ZiraHive Sync), AirPlay from the browser

- Remote access over Tailscale; public share links only through a separate address
- Offsite encrypted backup (restic); health and security checks; plugins (Telegram, webhook)
- Optional local AI: photo search, faces, Whisper subtitles, "Ask your files"
- Bangla and English interface

**Out of scope (for now)**

- Native Android, iOS or TV apps (a decided product choice: browser only)
- Hosting for other households on one server, or a paid hosted version
- macOS and Linux installers (the Docker stack runs there, but setup is Windows-only)
- Running without Docker (a native Windows installer is a future idea)
- Supplying IPTV content or subscriptions; DRM-protected streaming services
- Editing photos or videos (ZiraHive stores and plays them; clean-up only suggests what to delete)

## 6. Business requirements

Fifteen business requirements carry the objectives; all Must items are met in 1.0 or on `develop`.

| ID | Requirement | Objective | Priority | Status |
|---|---|---|---|---|
| BR-01 | Back up photos, videos and documents from phones and PCs to the home PC | O1, O2 | Must | Done (1.0) |
| BR-02 | Stream the family's movies, series, music and books to phone, PC and TV | O3 | Must | Done (1.0) |
| BR-03 | Use media where it already is, without copying it or sharing all drives | O3, O5 | Must | Done (1.0) |
| BR-04 | Reach it securely from outside the home, behind CGNAT, without opening a port | O5 | Must | Done (1.0) |

| ID | Requirement | Objective | Priority | Status |
|---|---|---|---|---|
| BR-05 | One account per family member, with storage limits and age limits for children | O2 | Must | Done (1.0) |
| BR-06 | Keep an encrypted copy of everything off the PC, and prove it can be restored | O4 | Must | Done (1.0) |
| BR-07 | A private space whose contents even the admin cannot read | O5 | Must | Done (1.0); photo vault and folders Done (develop) |
| BR-08 | Watch together from different places, in sync | O3 | Should | Done (1.0) |
| BR-09 | Watch an existing IPTV subscription inside ZiraHive | O3 | Should | Done (develop) |
| BR-10 | Use no paid service; everything free and open source | O7 | Must | Done (1.0) |
| BR-11 | Full Bangla interface | O8 | Must | Done (1.0) |
| BR-12 | Install on Windows with one guided script | O6 | Must | Done (1.0); untested on a second PC |
| BR-13 | Find documents and photos by content, without sending data out | O3, O5 | Could | Done (1.0, optional AI) |
| BR-14 | Let family with their own ZiraHive share categories with each other | O3 | Could | Done (1.0) |
| BR-15 | Extend it with add-ons (notifications, home automation) | O3 | Could | Done (1.0) |

# 7. Functional requirements

The system shall provide the functions below; each traces to a business requirement (BR) and is Done unless marked otherwise.

## 7.1 Accounts and access (BR-05)

| ID | The system shall | Status |
|---|---|---|
| FR-ACC-01 | Create the admin account on first run; let the admin add, edit and remove people | Done (1.0) |
| FR-ACC-02 | Set a storage quota and a role per account | Done (1.0) |
| FR-ACC-03 | Limit a child profile to titles rated 7+ or 13+, hide unrated titles and Live TV from it | Done (1.0) |
| FR-ACC-04 | Support passkeys (WebAuthn), TOTP two-factor sign-in with recovery codes, sign out everywhere | Done (1.0) |
| FR-ACC-05 | Throttle password guessing per address and keep an audit log for the admin | Done (1.0) |
| FR-ACC-06 | Let each account choose Bangla or English | Done (1.0) |

## 7.2 Backup, files and search (BR-01, BR-13)

| ID | The system shall | Status |
|---|---|---|
| FR-FIL-01 | Accept resumable chunked uploads with SHA-256 de-duplication; skip files already backed up | Done (1.0) |
| FR-FIL-02 | Store files under their real names in the data folder, by user and folder | Done (1.0) |
| FR-FIL-03 | Provide folders, rename, move, download with resume | Done (1.0) |
| FR-FIL-04 | Keep the last 20 versions of a document uploaded again under the same name | Done (1.0) |

| ID | The system shall | Status |
|---|---|---|
| FR-FIL-05 | Move deleted files and photos to a trash, restorable for 30 days, then purge | Done (1.0); photo delete Done (develop) |
| FR-FIL-06 | Index words inside PDFs, Office and text files; OCR scanned pages in Bangla and English | Done (1.0) |
| FR-FIL-07 | Scan paper with the phone camera into a searchable PDF | Done (1.0) |
| FR-FIL-08 | Sync a Windows folder both ways with ZiraHive Sync, with conflict and mass-delete guards | Done (1.0) |

## 7.3 Photos (BR-01)

| ID | The system shall | Status |
|---|---|---|
| FR-PHO-01 | Show a timeline by date taken (EXIF), with thumbnails and screen-sized previews, HEIC included | Done (1.0) |
| FR-PHO-02 | Offer albums, shared albums others can add to, and On this day memories | Done (1.0) |
| FR-PHO-03 | Suggest near-duplicates and blurry photos to clean up | Done (1.0) |
| FR-PHO-04 | Move photos to the trash from the viewer and from Select mode | Done (develop) |
| FR-PHO-05 | Move photos to the vault: encrypt, verify the vault copy byte for byte, then delete the original for good and clear the browser cache | Done (develop) |

## 7.4 Vault (BR-07)

| ID | The system shall | Status |
|---|---|---|
| FR-VLT-01 | Encrypt files, names and thumbnails in the browser before upload; the server stores ciphertext only | Done (1.0) |
| FR-VLT-02 | Unlock with a passphrase; issue a recovery key at set-up; allow passphrase change | Done (1.0) |
| FR-VLT-03 | Refuse vault requests for items of another account, admins included | Done (1.0) |
| FR-VLT-04 | Lock after inactivity: 5 minutes by default, adjustable 1–120 per browser | Done (develop) |
| FR-VLT-05 | Show the vault's photos and videos in Photos, by month, with a viewer | Done (develop) |
| FR-VLT-06 | Support nested folders with encrypted names: create, rename, move items, delete with contents | Done (develop) |
| FR-VLT-07 | Move a vault photo back to Photos with its original date, deleting the vault copy only after a confirmed upload | Done (develop) |
| FR-VLT-08 | Open only on https or localhost (secure context) | Done (1.0) |

## 7.5 Watch, listen, read (BR-02, BR-03)

| ID | The system shall | Status |
|---|---|---|
| FR-MED-01 | Turn each folder in the media folder into a category, read-only, typed from its name | Done (1.0) |
| FR-MED-02 | Recognise movies, series (S01E02 and similar), music tags, audiobooks, EPUB/PDF, comics | Done (1.0) |
| FR-MED-03 | Play directly when the browser can; otherwise convert to HLS while playing (NVENC or CPU), with quality, audio and subtitle choice | Done (1.0) |

| ID | The system shall | Status |
|---|---|---|
| FR-MED-04 | Remember position across devices; Continue watching, listening and reading | Done (1.0) |
| FR-MED-05 | Make offline copies (MP4) | Done (1.0) |
| FR-MED-06 | Cast to Chromecast; serve DLNA to TVs through ZiraHive Sync | Done (1.0) |

## 7.6 Live TV (BR-09)

| ID | The system shall | Status |
|---|---|---|
| FR-TV-01 | Add Xtream Codes sources; keep the provider password on the server only | Done (develop) |
| FR-TV-02 | List live channels by category, with search, logos and programme guide when the provider sends one | Done (develop) |
| FR-TV-03 | Play live channels as HLS, remuxing, or converting when the codec is not browser-friendly | Done (develop) |
| FR-TV-04 | Switch channel with arrow keys, Page Up/Down or a TV remote's channel keys | Done (develop) |
| FR-TV-05 | Browse provider movies and series with resume, and keep favourites | Done (develop); untested, provider has no VOD |

## 7.7 Together, sharing and remote (BR-04, BR-08, BR-14)

| ID | The system shall | Status |
|---|---|---|
| FR-SOC-01 | Run watch parties: scheduled or now, synced play over WebSocket, chat, reactions, voice (WebRTC) | Done (1.0) |
| FR-SOC-02 | Create share links for files, folders and albums with password, expiry, view-only | Done (1.0) |

| ID | The system shall | Status |
|---|---|---|
| FR-SOC-03 | Link two ZiraHive servers with invite codes for chosen categories and an age limit | Done (1.0) |
| FR-REM-01 | Serve the app remotely only over Tailscale; serve share links alone on an optional public address | Done (1.0) |

## 7.8 Operations (BR-06, BR-12, BR-15)

| ID | The system shall | Status |
|---|---|---|
| FR-OPS-01 | Install on Windows with `setup.ps1` (data and media folders, firewall, shortcut) and update with `update.ps1` | Done (1.0) |
| FR-OPS-02 | Back up photos, files, vault and database daily with restic to disk, S3, SFTP or rclone; check weekly | Done (1.0) |
| FR-OPS-03 | Show health and security checks in Settings and through `python -m app.doctor` | Done (1.0) |
| FR-OPS-04 | Upgrade the database with Alembic migrations on start | Done (1.0) |
| FR-OPS-05 | Load plugins (Telegram, signed webhook, custom Python) only when turned on | Done (1.0) |
| FR-OPS-06 | Notify in the app and through ntfy (backups, albums, new videos, disk 90% full) | Done (1.0) |
| FR-UI-01 | Collapse the side menu to icons, remembered per browser; phones keep the bottom bar | Done (develop) |

## 7.9 Local AI, optional (BR-13)

| ID | The system shall | Status |
|---|---|---|
| FR-AI-01 | Search photos by description (CLIP), Bangla queries translated locally first | Done (1.0) |

| ID | The system shall | Status |
|---|---|---|
| FR-AI-02 | Group faces into People, with names, merge and "not this person" | Done (1.0) |
| FR-AI-03 | Generate subtitles with Whisper | Done (1.0) |
| FR-AI-04 | Answer questions from the user's own documents with numbered sources (Ask) | Done (1.0) |
| FR-AI-05 | Run entirely on the PC; degrade with a clear message when the AI profile is off | Done (1.0) |

## 8. Non-functional requirements

Security and privacy come first, then never losing data, then ease of use on modest home hardware.

| ID | Area | Requirement |
|---|---|---|
| NFR-01 | Security | The app is reachable only on the home network and through Tailscale; no port is opened on the router. Share links may use a separate public address that serves nothing else. |
| NFR-02 | Security | Security headers and a strict Content-Security-Policy; secure cookies; URL tokens only on media links, scoped to one title for 12 hours. |
| NFR-03 | Privacy | Vault content, names, thumbnails and folder layout are encrypted in the browser (PBKDF2-SHA256 600,000 iterations, AES-256-GCM, per-file keys, authenticated chunks). The server never receives the passphrase or keys. |
| NFR-04 | Privacy | No telemetry. Data leaves the PC only for features the admin turns on: offsite backup (encrypted), TMDB lookups, IPTV, Tailscale. |
| NFR-05 | Integrity | Media folders are mounted read-only; ZiraHive cannot change or delete them. |

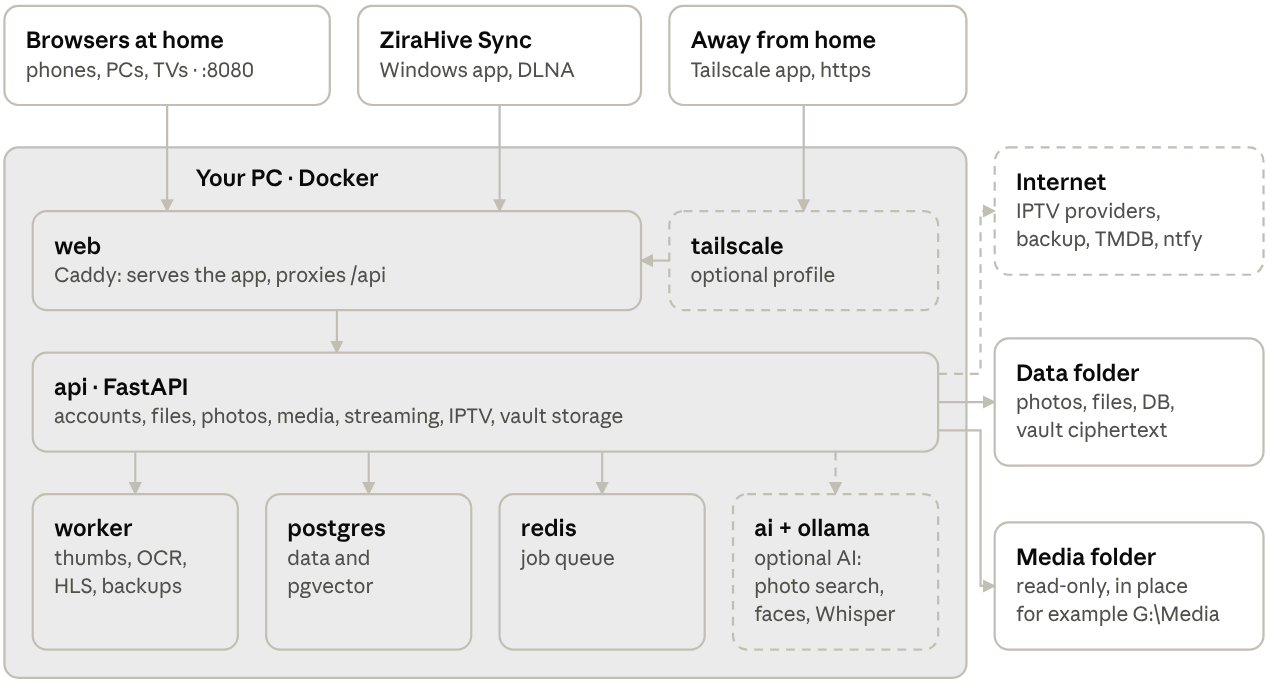
| ID | Area | Requirement |
|---|---|---|
| NFR-06 | Reliability | Uploads resume after a dropped connection; every upload is checked by SHA-256. |
| NFR-07 | Recoverability | Daily offsite snapshot with a weekly read-back check; restore guide in `docs/BACKUP.md`; data folder readable in Explorer without ZiraHive. |
| NFR-08 | Performance | A library of several thousand titles browses without paging delays; one 1080p conversion runs in real time on a recent 6-core CPU, faster with NVENC; a limit on simultaneous conversions protects the PC. |
| NFR-09 | Capacity | Sized for one household: about 2–10 accounts, a few concurrent streams, tens of thousands of photos. |
| NFR-10 | Usability | Every function works from a phone browser and is installable to the home screen; no store apps. |
| NFR-11 | Localisation | Every interface text has a Bangla translation, with Bangla digits and dates; CI fails on a missing one. |
| NFR-12 | Accessibility | Keyboard use throughout (including Live TV channel keys), labelled controls, dialogs that focus their first field. |
| NFR-13 | Portability | Runs anywhere Docker runs; the guided installer targets Windows 10/11 with Docker Desktop (WSL 2). |
| NFR-14 | Maintainability | API tests (pytest), Playwright browser tests, Alembic migrations with a test that fails when models change without one, GitHub Actions CI. |
| NFR-15 | Cost | No paid dependency. Optional services (TMDB key, IPTV subscription, Tailscale free plan up to 6 users) are supplied by the user. |

NFR-08 and NFR-09 are design targets, not yet load-tested on the owner's hardware; the Locust load test exists to measure them.

# 9. System architecture, components and data

ZiraHive is one Docker Compose stack on the home PC; every client talks to the `web` container, and only the `api` touches the folders and the internet.

**One Windows PC runs every part; browsers are the only clients**




ZiraHive architecture · 3 clients, 7 containers, 2 folders

Dashed parts are optional: `tailscale` runs with `--profile remote`, `ai` and `ollama` with `--profile ai`, and outside services only when the admin sets them up.

| Component | Technology | Responsibility |
|---|---|---|
| `web` | Caddy, React + TypeScript (Vite) | Serves the single-page app, security headers, proxies `/api` |
| `api` | Python, FastAPI, SQLAlchemy, Alembic | Accounts, uploads, files, photos, media scan and streaming, IPTV, parties (WebSocket), vault storage, plugins |
| `worker` | Python job runner on Redis | Thumbnails, previews, library scans, OCR, HLS conversion, offline copies, offsite backup, nightly clean-up |

| Component | Technology | Responsibility |
|---|---|---|
| `postgres` | PostgreSQL + pgvector | All metadata; vectors for AI search |
| `redis` | Redis | Job queue between `api` and `worker` |
| `tailscale` | Tailscale (optional) | Private HTTPS address on the tailnet; optional Funnel for share links |
| `ai`, `ollama` | fastembed CLIP, YuNet + SFace, faster-whisper; Qwen2.5 3B, BGE-M3 (optional) | Photo search, faces, subtitles, Ask your files |

### Where data lives

| Data | Location | Notes |
|---|---|---|
| Backed-up photos and files | Data folder, `storage/<user>/<folder>/` | Real file names; browsable in Explorer |
| Thumbnails, previews, HLS segments | Data folder caches | Rebuilt when missing |
| Vault items | Data folder, `vault/<user>/<id>.bin` and `.thumb` | Ciphertext only; names and folders inside encrypted metadata |
| Database | A Docker volume for PostgreSQL | Accounts, assets, albums, shares, libraries, parties, IPTV sources, audit log |
| Media | Media folder (and any listed drives), mounted read-only | Never written |
| Offsite snapshots | restic repository (disk, S3, SFTP, rclone) | Encrypted; include a database dump |

Main data entities: User, Asset (with versions, extracted text, faces and photo vectors), Album, Share link, Library, Media item and Series, Party, Vault and Vault item, IPTV source and favourite, Plugin settings, Audit event. Schema changes ship as Alembic migrations, checked by a test against both SQLite and PostgreSQL.

# 10. External interfaces and integrations

Every outside service is optional and user-supplied; ZiraHive works at home with none of them.

| Interface | Direction | Used for | Required? |
|---|---|---|---|
| Web app over HTTP(S) | Browser → PC | The whole user interface and REST API ( `/api` , OpenAPI docs at `/docs` in development) | Yes |
| WebSocket | Browser ↔ PC | Watch-party sync, chat, voice signalling | For parties |
| Tailscale | PC ↔ tailnet | Remote access with HTTPS (MagicDNS); optional Funnel address for share links | For remote use |
| Xtream Codes API | PC → IPTV provider | Categories, channels, guide, VOD; streams remuxed by the PC | For Live TV |
| restic repository | PC → disk, S3, SFTP, rclone | Encrypted offsite snapshots | Recommended |
| TMDB API | PC → themoviedb.org | Posters, plot summaries, age certificates | No |
| ntfy | PC → ntfy server | Phone notifications | No |
| Telegram, webhooks | PC → chat or automation | Plugin events (new media, backup failure, party started) | No |
| Chromecast (Cast SDK), AirPlay | Browser → TV | Casting; TV fetches one title with a 12-hour token | No |
| DLNA / UPnP | ZiraHive Sync → TVs on the LAN | Browsing and playing from a TV's media menu | No |
| ZiraHive Sync API | Windows app ↔ PC | Two-way folder sync, device sign-in | No |

| Interface | Direction | Used for | Required? |
|---|---|---|---|
| Linked ZiraHive server | Server ↔ server | Shared categories, proxied streaming, revocable tokens | No |
| Ollama, AI container | api → local containers | CLIP search, faces, Whisper, Ask (Qwen2.5 3B, BGE-M3) | No ( `--profile ai` ) |

## 11. Constraints, assumptions and dependencies

ZiraHive depends on one always-on Windows PC with Docker Desktop; everything else follows from that.

**Constraints**

- One PC is the whole server: when it sleeps or is off, ZiraHive is unavailable (no failover).
- Browser-only clients: background phone backup is limited by what mobile browsers allow (uploads pause when the screen locks and resume when ZiraHive is opened again).
- iPhone Safari may hand over JPEG copies of HEIC photos and only the still part of Live Photos.
- The vault needs a secure context (https or localhost); on plain LAN http it refuses to open.
- Windows networking: with WSL mirrored networking, phones need both a Windows Firewall rule and a Hyper-V firewall rule, and the Wi-Fi profile must be Private.
- Licence AGPL-3.0: anyone offering a modified version over a network must publish its source.

**Assumptions**

- The PC has free disk space for backups (setup picks the drive with the most) and is on most of the day.
- Each household has one technically confident admin.
- Family members join the tailnet (Tailscale app) to use ZiraHive away from home.
- IPTV sources are subscriptions the user is entitled to use.

**Dependencies**

| Dependency | Role |
|---|---|
| Docker Desktop (WSL 2) | Runs the containers on Windows |
| PostgreSQL with pgvector, Redis | Database and vector search; job queue |
| Caddy | Web server and reverse proxy |
| FastAPI, SQLAlchemy, Alembic | API, data access, migrations |
| React, TypeScript, Vite, hls.js, epub.js, pdf.js | Web app and players |
| ffmpeg / ffprobe | Media probing, thumbnails, HLS conversion, Live TV remux |
| Tesseract | OCR (Bangla and English) |
| restic, rclone | Offsite backup |
| Tailscale | Remote access |
| fastembed CLIP, OpenCV YuNet + SFace, faster-whisper, Ollama | Optional local AI |

## 12. Risks and mitigations

The two highest risks are losing data when the single PC fails and leaking a secret through the repository; the second already happened once.

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| PC disk fails or the PC is stolen | Medium | High | Daily restic offsite backup with weekly read-back; restore guide; data folder readable without ZiraHive |
| A secret is committed to git (the vault recovery key was pushed on `test_branch` on 18 Oct 2026) | Happened | High | Key to be replaced (vault was empty); branch rewritten locally, force-push pending; `*recovery-key*.txt` and `.env` now ignored |

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Forgotten vault passphrase and lost recovery key | Medium | High (vault unrecoverable) | Recovery key shown at set-up with Copy and Save; guidance to keep it in a password manager |
| A weak vault passphrase is guessed offline from a disk copy | Low | High | 600,000 PBKDF2 iterations; minimum 10 characters; advice to use a long phrase |
| Plaintext traces of photos moved into the vault (disk sectors, older backups) | Medium | Medium | Add private photos straight into the vault; older snapshots expire by retention |
| Exposing the app to the internet by mistake | Low | High | Tailscale-only remote access; gateway guard; security check panel |
| IPTV provider blocks or limits connections; poor or missing guide data | Medium | Low | One provider connection per channel shared by all viewers; guide shown only when sent |
| Setup fails on another PC (Docker, WSL, firewall, Public network profile) | Medium | Medium | `setup.ps1` checks and explains each step; FIRST_RUN guide; health page; `app.doctor` |
| Conversions overload the PC | Medium | Medium | Limit on simultaneous conversions; NVENC when present; idle stop |
| One maintainer (bus factor 1) | High | Medium | Open source, documented, tested, CI on every push |

## 13. Release status, acceptance criteria and test approach

Release 1.0 is shipped; 1.1 is ready to release once the open items below pass.

| Date | Release or milestone | Contents |
|---|---|---|
| 18 Oct 2026 | `develop` (toward 1.1) | Live TV (IPTV); photo trash; photo vault with Move to vault; vault folders; auto-lock 5 min adjustable; collapsible side menu; dialog focus and Live TV fixes; test helpers in `test-zh/` |
| 17 Oct 2026 | Setup fixes | `setup.ps1` checks 127.0.0.1, shows wait times, opens the WSL firewall for phones |
| 17 Oct 2026 | 1.0.0 ( `main` , tag `v1.0.0` ) | All nine planned batches |

### Acceptance criteria for 1.1

- [ ] All API tests and Playwright suites pass in CI on PostgreSQL
- [x] ~~Live TV: channels play, keys switch channel, layout fits from phone to desktop (tested on the owner's PC, 18 Oct)~~
- [x] ~~Photo vault: move in, verify, view, move back, add from device, delete, folders, auto-lock (tested 18 Oct)~~
- [ ] Live proof that a second account gets "not found" for another account's vault items
- [ ] Renaming a vault folder keeps the view inside it (fixed, needs re-test with the vault unlocked)
- [ ] Vault recovery key replaced and `test_branch` force-pushed without it
- [ ] Live TV movies and series checked with a provider that offers VOD
- [ ] Fresh install on a second Windows PC within 30 minutes (O6)

### Test approach

| Level | Tool | What it covers |
|---|---|---|
| Unit and API | pytest ( `server/tests` ), real ffmpeg on synthetic streams | Endpoints, migrations, IPTV parsing and remux |
| Browser end-to-end | Playwright ( `e2e/tests` ), Chrome for H.264 | Sign-in, uploads, players, parties (two browsers), CSP |
| Load | Locust ( `loadtest/` ) | Concurrent browsing and streaming |

| Level | Tool | What it covers |
|---|---|---|
| On the real PC | Built-in browser and `test-zh/` helpers | Rebuilds, setup runs, network and firewall checks, manual feature passes |
| Localisation | `npm run check:i18n` in CI | No text without Bangla |

The tests still to run later are listed in the project note `claude/test-backlog.md`.

## 14. Future expansion ideas and roadmap

The next step is to make ZiraHive installable by other families; after that, the biggest gains are removing the single-PC risk and a TV-first interface.

**Next (1.1–1.2): finish and harden**

| Idea | Why it matters |
|---|---|
| Close the 1.1 acceptance list (section 13) and release 1.1 from `develop` | Ships Live TV and the photo vault to `main` |
| Install test on a second Windows PC; fix what it finds | Proves objective O6 before inviting other families |
| Move to vault straight into a chosen folder; Move to vault from Files | Fewer steps for private documents |
| Live TV: XMLTV guide URL, M3U playlists, catch-up for channels that offer archive | Fills the missing guide; supports more providers |
| Remux instead of re-encode when only the audio needs converting; burned-in picture subtitles (PGS) | Less CPU, more files play |
| Converted streams for DLNA TVs that can't play a file | Every file plays on every TV |

**Later (1.3–2.0): grow beyond one PC**

| Idea | Why it matters |
|---|---|
| Native Windows installer without Docker (bundled Python, ffmpeg, PostgreSQL as a service) | Removes the hardest install step |
| Back up to a relative's ZiraHive over a linked-server connection (encrypted restic repository) | Free offsite backup between families |
| Standby replica on a second PC, or a NAS/Raspberry Pi (ARM) image | Removes the single point of failure |
| TV-first "10-foot" interface navigated with a remote | Better living-room use than the phone layout |
| Live TV recording into the media folder; scheduled recordings from the guide | DVR-style use of the subscription |
| Kids' screen-time limits and a bedtime lock | Extends the age-limit feature parents already use |
| Wake-on-LAN and a sleep schedule for the PC | Saves power while staying reachable |
| Sharing vault items between family members (public-key wrapped file keys) | Private sharing without the admin seeing it |

### Exploring: smarter and wider

| Idea | Why it matters |
|---|---|
| Better Bangla photo search with a multilingual picture model small enough for CPUs | Natural Bangla queries without translation |
| Automatic albums for trips and events; video scene search | Finds memories without manual albums |
| Document assistant: recognise bills, IDs and certificates; remind before a passport or licence expires | Turns the file store into household admin help |

| Idea | Why it matters |
|---|---|
| Home Assistant integration (presence, "movie mode" lights) beyond the webhook plugin | Fits into smart homes |
| Community plugin catalogue | Others extend ZiraHive without changing its core |
| macOS and Linux installers | More households can run it |
| Optional managed relay or hosted set-up service for non-technical families | A possible way to fund the project without a paywall on features |

Native phone apps stay out of scope unless browser limits on background backup become the main complaint.

## 15. Glossary

| Term | Meaning |
|---|---|
| BRD / SRS | Business Requirements Document / Software Requirements Specification |
| CGNAT | Carrier-grade NAT: the ISP shares one public address among many homes, so incoming connections can't be forwarded |
| Category | A folder in the media folder that ZiraHive shows as a library (Movies, Series, Music…) |
| Data folder | `ZIRAHIVE_DATA_PATH`: where backups, files, the vault and the database live (on the owner's PC, `D:\GharBoxData`) |
| Media folder | `MEDIA_PATH`: the read-only folder of films, music and books (on the owner's PC, `G:\Media`) |
| DLNA / UPnP | Standards smart TVs use to browse and play media on the home network |
| HLS | HTTP Live Streaming: video cut into short segments a browser can play while it is converted |

| Term | Meaning |
|---|---|
| NVENC | NVIDIA's hardware video encoder |
| Remux | Repackaging a stream into another container without re-encoding it |
| Tailscale, tailnet | A private encrypted network of your own devices; ZiraHive's address on it is `https://zirahive.<tailnet>.ts.net` |
| Xtream Codes | The common login format of IPTV providers: server address, username, password |
| EPG | Electronic programme guide: what is on each channel now and next |
| VOD | Video on demand: an IPTV provider's movies and series |
| Vault | The end-to-end encrypted space; unlocked with a passphrase, recoverable only with the recovery key |
| Recovery key | 32 random bytes shown once at vault set-up; the only way in without the passphrase |
| Secure context | A page served over https or localhost, where browsers allow Web Crypto |
| restic | The encrypted, deduplicating backup tool used for offsite copies |
| PWA | Progressive web app: a website added to the home screen that opens like an app |
| AGPL-3.0 | The open-source licence: modified versions offered over a network must publish their source |